

```

mysql> /* Queries and Sub-Queries
/*> Sub-Query is a query inside another query.
/*> It is written inside paranthesis() and its result is used by the outer
query.
/*> It is also called as inner query, nested query, inner select
/*>
/*> purpose of Sub-Query
/*> 1. to breakdown complex problems into smaller queries.
/*> 2. to use the result of one query as input to another.
/*> 3. to avoid multiple steps (write in a single query instead of temp tables).
/*>
/*> general syntax
/*> select column_list from table where column operator(select column from table
where condition);
/*>
/*> types of Sub-Queries
/*> 1. single row sub-query
/*> 2. multi row sub-query
/*> 3. nested sub-query
/*> 4. correlated sub-query
/*> 5. exists/not exists sub-query
/*>
/*> single row sub-query: single row subqueries returns exactly one row and one
column. They use comparision operators like =,>,<,<=,>=,!=
/*> characteristics:
/*> > returns only one value
/*> > can use single-value comparision operators.
/*> > if subquery returns no rows or multiple rows, it causes an error.
/*>
/*> syntax: select columns from table where column operator (select column from
table where condition);
/*>
/*> multi row sub-query:
/*> a. multi row sub-query in: it returns more than one row. the in operator is
used when you want to compare a value with multiple values returned by the subquery
/*> characteristics:
/*> > returns multiple rows.
/*> > use the in operator.
/*> > checks whether a value exists in the list returned by the subquery.
/*> > equivalent to multiple or condition.
/*> > most commonly used for matching values.
/*>
/*> suntax: select column from table where column in(select column from table
where condition);
/*>
/*> b. multi row subquery any/some: it returns multiple rows when using any or
some. The condition is true if it matches atleast one value returned by the
subquery.
/*> characteristics:
/*> > returns multiple values.

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/*> > any and some are interchangeable.
/*> > condition is true if any of the values satisfy the condition.
/*> > often used with in, >any, <any, >=any ... etc
/*>
/*> syntax: select column from table where column operator any/some(select
column from table where condition);
/*>
/*> c. multi row subquery all: The all operator means the condition must be true
for all values returned by the subquery.
/*> characteristics:
/*> > conditions must be true for all values in the subquery result.
/*> > often used with >all, <all, <=all, >= all
/*> > >all means greater than maximum value
/*> > <all means less than minimum value
/*>
/*> syntax: select column from table where column operator all(select column
from table where condition);
/*>
/*> nested sub-query: They are subqueries within subqueries. they can be
multiple levels deep and are executed from innermost to the outermost.
/*> characteristics:
/*> > multiple levels of nesting
/*> > inner queries execute first.
/*> > can combine different types of subqueries
/*> > useful for complex conditions
/*>
/*> syntax: select column_list from table where column = (select column from
table where column = (select column from table where condition));
/*>
/*> correlated sub-query: they are subqueries that reference columns from outer
query. unlike regular subqueries that execute once. correlated subqueries execute
once per each row processed by the outerquery
/*> characteristics:
/*> > reference column from outer query.
/*> > executes multiple times(once per outer row).
/*> > cannot be executed independently.
/*> > often used with exists/not exists.
/*> > generally slower than non-correlated subquerues.
/*>
/*> syntax: select column from table1 outer_alias where condition(select column
from table2 inner_alias where inner_alias.column = outer_alias.column);
/*>
/*> exists/not exists sub-query: exists is used to test weather a subquery
returns any rows. it returns TRUE if the subquery returns as least one row. False
otherwise.
/*>
not exists return TRUE if the subquery returns
no rows.
/*> characteristics:
/*> > returns TRUE/FALSE(boolean result).
/*> > more efficient than in for large datasets.

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/*> > often used with correlated subqueries.
/*> > stops execution as soon as first match is found.
/*> > select clauses in exists doesn't matter(often use select 1).
/*>
/*> syntax: select columns from table1 where exists(select 1 from table2 where
condition).
/*> syntax: select columns from table1 where not exists(select 1 from table2
where condition). */
mysql> drop table products;
ERROR 1046 (3D000): No database selected
mysql> use cg_mysql;
Database changed
mysql> drop table products;
Query OK, 0 rows affected (0.10 sec)
```

```
mysql> drop table orders;
Query OK, 0 rows affected (0.09 sec)
```

```
mysql> -- Step 1: Create the tables
mysql> CREATE TABLE Customers (
->   customer_id    INT PRIMARY KEY,
->   customer_name  VARCHAR(50),
->   city           VARCHAR(50),
->   customer_type  VARCHAR(20)
-> );
Query OK, 0 rows affected (0.06 sec)
```

```
mysql>
mysql> CREATE TABLE Products (
->   product_id     INT PRIMARY KEY,
->   product_name   VARCHAR(50),
->   category       VARCHAR(30),
->   price          DECIMAL(10,2)
-> );
Query OK, 0 rows affected (0.06 sec)
```

```
mysql>
mysql> CREATE TABLE Orders (
->   order_id       INT PRIMARY KEY,
->   customer_id    INT REFERENCES Customers(customer_id),
->   product_id     INT REFERENCES Products(product_id),
->   quantity       INT,
->   order_date     DATE,
->   amount         DECIMAL(10,2)
-> );
Query OK, 0 rows affected (0.05 sec)
```

```
mysql>
mysql> -- Step 2: Insert Customers records
mysql> INSERT INTO Customers (customer_id, customer_name, city, customer_type)
```

```
VALUES (1, 'Rahul', 'Delhi', 'Premium');  
Query OK, 1 row affected (0.02 sec)
```

```
mysql> INSERT INTO Customers (customer_id, customer_name, city, customer_type)  
VALUES (2, 'Meena', 'Mumbai', 'Regular');  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Customers (customer_id, customer_name, city, customer_type)  
VALUES (3, 'Arjun', 'Bangalore', 'Premium');  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Customers (customer_id, customer_name, city, customer_type)  
VALUES (4, 'Kavya', 'Delhi', 'Regular');  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Customers (customer_id, customer_name, city, customer_type)  
VALUES (5, 'Suresh', 'Chennai', 'Premium');  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Customers (customer_id, customer_name, city, customer_type)  
VALUES (6, 'Anjali', 'Mumbai', 'Regular');  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Customers (customer_id, customer_name, city, customer_type)  
VALUES (7, 'Vikram', 'Delhi', 'Regular');  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Customers (customer_id, customer_name, city, customer_type)  
VALUES (8, 'Pooja', 'Bangalore', 'Premium');  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Customers (customer_id, customer_name, city, customer_type)  
VALUES (9, 'Devi', 'Chennai', 'Regular');  
Query OK, 1 row affected (0.01 sec)
```

```
mysql>
```

```
mysql> -- Step 3: Insert Products records
```

```
mysql> INSERT INTO Products (product_id, product_name, category, price) VALUES (1,  
'Laptop', 'Electronics', 55000);  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Products (product_id, product_name, category, price) VALUES (2,  
'Mobile', 'Electronics', 25000);  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Products (product_id, product_name, category, price) VALUES (3,  
'Chair', 'Furniture', 7000);  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Products (product_id, product_name, category, price) VALUES (4,
```

```
'Table', 'Furniture', 9000);  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Products (product_id, product_name, category, price) VALUES (5,  
'Headphones', 'Electronics', 3000);  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Products (product_id, product_name, category, price) VALUES (6,  
'Bookshelf', 'Furniture', 6000);  
Query OK, 1 row affected (0.01 sec)
```

```
mysql>  
mysql> -- Step 4: Insert Orders records  
mysql> INSERT INTO Orders (order_id, customer_id, product_id, quantity, order_date,  
amount) VALUES (1, 1, 1, 1, '2023-01-05', 55000);  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Orders (order_id, customer_id, product_id, quantity, order_date,  
amount) VALUES (2, 2, 2, 1, '2023-01-10', 25000);  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Orders (order_id, customer_id, product_id, quantity, order_date,  
amount) VALUES (3, 3, 3, 2, '2023-02-01', 14000);  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Orders (order_id, customer_id, product_id, quantity, order_date,  
amount) VALUES (4, 1, 5, 2, '2023-02-15', 6000);  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Orders (order_id, customer_id, product_id, quantity, order_date,  
amount) VALUES (5, 4, 4, 1, '2023-03-03', 9000);  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Orders (order_id, customer_id, product_id, quantity, order_date,  
amount) VALUES (6, 5, 2, 2, '2023-03-20', 50000);  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Orders (order_id, customer_id, product_id, quantity, order_date,  
amount) VALUES (7, 6, 4, 3, '2023-04-02', 18000);  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Orders (order_id, customer_id, product_id, quantity, order_date,  
amount) VALUES (8, 3, 5, 1, '2023-04-15', 3000);  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Orders (order_id, customer_id, product_id, quantity, order_date,  
amount) VALUES (9, 7, 3, 1, '2023-05-01', 7000);  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Orders (order_id, customer_id, product_id, quantity, order_date,
```

```
amount) VALUES (10, 8, 1, 1, '2023-05-20', 54000);
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Orders (order_id, customer_id, product_id, quantity, order_date,
amount) VALUES (11, 2, 3, 2, '2023-06-02', 12000);
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Orders (order_id, customer_id, product_id, quantity, order_date,
amount) VALUES (12, 5, 4, 1, '2023-06-15', 8500);
Query OK, 1 row affected (0.03 sec)
```

```
mysql> select * from customers;
```

customer_id	customer_name	city	customer_type
1	Rahul	Delhi	Premium
2	Meena	Mumbai	Regular
3	Arjun	Bangalore	Premium
4	Kavya	Delhi	Regular
5	Suresh	Chennai	Premium
6	Anjali	Mumbai	Regular
7	Vikram	Delhi	Regular
8	Pooja	Bangalore	Premium
9	Devi	Chennai	Regular

```
9 rows in set (0.05 sec)
```

```
mysql> select * from products;
```

product_id	product_name	category	price
1	Laptop	Electronics	55000.00
2	Mobile	Electronics	25000.00
3	Chair	Furniture	7000.00
4	Table	Furniture	9000.00
5	Headphones	Electronics	3000.00
6	Bookshelf	Furniture	6000.00

```
6 rows in set (0.05 sec)
```

```
mysql> select * from orders;
```

order_id	customer_id	product_id	quantity	order_date	amount
1	1	1	1	2023-01-05	55000.00
2	2	2	1	2023-01-10	25000.00
3	3	3	2	2023-02-01	14000.00
4	1	5	2	2023-02-15	6000.00
5	4	4	1	2023-03-03	9000.00
6	5	2	2	2023-03-20	50000.00

7	6	4	3	2023-04-02	18000.00
8	3	5	1	2023-04-15	3000.00
9	7	3	1	2023-05-01	7000.00
10	8	1	1	2023-05-20	54000.00
11	2	3	2	2023-06-02	12000.00
12	5	4	1	2023-06-15	8500.00

12 rows in set (0.00 sec)

```
mysql> -- find the name and price of the most expensive product.
mysql> select product_name, price from products where price = (select max(price)
from products);
```

product_name	price
Laptop	55000.00

1 row in set (0.06 sec)

```
mysql> select order_id, customer_id, amount from orders where amount = (select
min(amount) from orders);
```

order_id	customer_id	amount
8	3	3000.00

1 row in set (0.00 sec)

```
mysql> select customer_name from customer where customer_id = (select customer_id
from orders where order_id = 5);
```

ERROR 1146 (42S02): Table 'cg_mysql.customer' doesn't exist

```
mysql> select customer_name from customers where customer_id = (select customer_id
from orders where order_id = 5);
```

customer_name
Kavya

1 row in set (0.04 sec)

```
mysql> select category from products where product_id = (select product_id from
orders where order_id = 1);
```

category
Electronics

1 row in set (0.00 sec)

```
mysql> select product_name from products where product_id = (select product_id from
```

```
orders where amount = (select max(amount) from orders));
```

```
+-----+
| product_name |
+-----+
| Laptop       |
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select customer_name, city from customers where city in ('Delhi','Mumbai');
```

```
+-----+-----+
| customer_name | city |
+-----+-----+
| Rahul         | Delhi |
| Meena         | Mumbai |
| Kavya         | Delhi |
| Anjali        | Mumbai |
| Vikram        | Delhi |
+-----+-----+
```

```
5 rows in set (0.03 sec)
```

```
mysql> select product_name from products where product_id in (select product_id
from orders);
```

```
+-----+
| product_name |
+-----+
| Laptop       |
| Mobile       |
| Chair        |
| Table        |
| Headphones   |
+-----+
```

```
5 rows in set (0.00 sec)
```

```
mysql> select customer_name from customers where customer_id in (select customer_id
from orders);
```

```
+-----+
| customer_name |
+-----+
| Rahul         |
| Meena         |
| Arjun         |
| Kavya         |
| Suresh       |
| Anjali        |
| Vikram        |
| Pooja        |
+-----+
```

```
8 rows in set (0.00 sec)
```

```
mysql> select product_name from products where product_id not in (select product_id
```

```
from orders);
```

```
+-----+  
| product_name |  
+-----+  
| Bookshelf    |  
+-----+
```

```
1 row in set (0.03 sec)
```

```
mysql> select customer_name from customers where city in (select city from  
customers where customer_type in ('premium'));
```

```
+-----+  
| customer_name |  
+-----+  
| Rahul         |  
| Arjun         |  
| Kavya         |  
| Suresh        |  
| Vikram        |  
| Pooja         |  
| Devi          |  
+-----+
```

```
7 rows in set (0.04 sec)
```

```
mysql> select product_name, price from products where price > any(select price from  
products where category = 'furniture');
```

```
+-----+-----+  
| product_name | price  |  
+-----+-----+  
| Laptop       | 55000.00 |  
| Mobile       | 25000.00 |  
| Chair        | 7000.00  |  
| Table        | 9000.00  |  
+-----+-----+
```

```
4 rows in set (0.00 sec)
```

```
mysql> select product_name, price from products where price < any(select price from  
products where category = 'electronics');
```

```
+-----+-----+  
| product_name | price  |  
+-----+-----+  
| Mobile       | 25000.00 |  
| Chair        | 7000.00  |  
| Table        | 9000.00  |  
| Headphones   | 3000.00  |  
| Bookshelf    | 6000.00  |  
+-----+-----+
```

```
5 rows in set (0.00 sec)
```

```
mysql> select order_id, amount from orders where amount > any (select amount from  
orders where customer_id = (select customer_id from customers where customer_name =
```

```
'Arjun'));
```

order_id	amount
1	55000.00
2	25000.00
3	14000.00
4	6000.00
5	9000.00
6	50000.00
7	18000.00
9	7000.00
10	54000.00
11	12000.00
12	8500.00

```
11 rows in set (0.00 sec)
```

```
mysql> select order_id, amount from orders where amount < any (select amount from
orders where customer_id = (select customer_id from customers where city =
'Mumbai'));
```

```
ERROR 1242 (21000): Subquery returns more than 1 row
```

```
mysql> select order_id, amount from orders where amount < any (select amount from
orders where customer_id in (select customer_id from customers where city =
'Mumbai'));
```

order_id	amount
3	14000.00
4	6000.00
5	9000.00
7	18000.00
8	3000.00
9	7000.00
11	12000.00
12	8500.00

```
8 rows in set (0.00 sec)
```

```
mysql> notee
```